



Anhui Moon Technology Co., Ltd.

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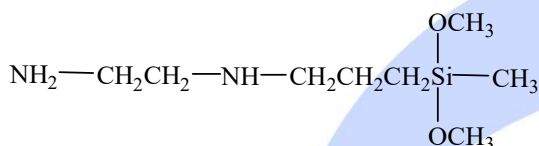
Product information

KH-602

N-(β-amino ethyl) -γ- amino propyl methyl dimethoxy silane

Product description

Structural formula:



Empirical formula: $\text{C}_8\text{H}_{22}\text{N}_2\text{O}_2\text{Si}$

Molecular weight: 206

CAS No.: 3069-29-2

Chemical name:

N-(β-amino ethyl)-γ-amino propyl methyl dimethoxy silane

Properties

KH-602 is a bifunctional organosilane possessing reactive primary amine and hydrolysable inorganic methoxysilyl groups. The dual nature of its reactivity allows it to bind chemically to both inorganic materials (e.g. glass, metals, fillers) and organic polymers (e.g. thermoplastics, thermosets, elastomers) thus functioning as an adhesion promoter, surface modifier and as a reactant for product modification.

KH-602 is a colorless or clear yellow liquid with an amine-like odor. It is soluble in alcohols and aliphatic or aromatic hydrocarbons.

Technical data

Typical characteristics	Value
Appearance	Colorless or clear yellow liquid
Purity	≥97%
Density at 25°C	0.968g/ml
Boiling point at 760mmHg	265°C
Flash point (closed cup)	>100°C
Refractive index (25°C)	1.445

Note: the above data are for reference only, can not be used as a technical specification

Reactivity

In the presence of water, the methoxy groups of KH-602 hydrolyze to form reactive silanol groups which can bond to a variety of inorganic substrates. The organophilic diamino group of KH-602 can react with a suitable polymer.

The hydrolysis takes place autocatalytically and the pH of the hydrolysate is about 10~11. Examples of suitable inorganic substrates are glass, glass fibers, glass wool, mineral wool, silicic acid, sand, cristobalite, wollastonite, mica and aluminum hydroxide, kaolin, talc, other silicate fillers, metal oxides and metals.

KH-602 may be used with such polymers as phenolic, furan resins and silicones.

KH-602 can undergo reactions with ketone or ester solvents. Silane or silanized substrates can react with carbon dioxide to form the corresponding carbonates and/or carbamates, respectively. Product modifications are possible through co-condensation with polysiloxanes (textile auxiliaries).

Application and performance

KH-602 can be used in many application. Examples include:

- ✧ Additive for cold-curing phenolic and furan foundry resins to improve the flexural strength of the



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sand-resin. Composites while maintaining the shelf life of the resin over long periods.

from **MOON TECH** supplier.

- ✧ Starting material in the synthesis of aminofunctional silicones.
- ✧ Additive to phenolic resin binders for mineral fiber insulating materials.
- ✧ Primer for additive to sealants and adhesives.
- ✧ Additive and/or primer for improving the adhesion of paints and coatings to the substrate.
- ✧ Sealants and adhesives: as a primer or additive and for chemical modification.
- ✧ Mineral-filled composites: for pretreatment of fillers and pigments.
- ✧ Paints and coatings: as an additive and primer for improving adhesion to the substrate.

Important product effects that can be achieved through the use of LT602 include:

Improved mechanical properties, such as strength, impact strength and modulus of elasticity.

- ✧ Improved moisture and corrosion resistance.
- ✧ Improved electrical properties, such as dielectric constant, volume resistivity.

KH-602 can also improve such processing properties as:

- ✧ Adhesion

The data presented in this leaflet are in accordance with the present state of our knowledge, but do not absolve the user from carefully checking all supplies immediately on receipt. We reserve the right to alter product constants within the scope of technical progress or new developments. The recommendations made in this leaflet should be checked by preliminary trials because of conditions during processing over which we have no control, especially where other companies' raw materials are also being used. The recommendations do not absolve the user from the obligation of investigating the possibility of infringement of third parties' rights and, if necessary, clarifying the position. Recommendations for use do not constitute a warranty, either express or implied, of the fitness or suitability of the products for a particular purpose.

Product safety, handling and storage

Customers considering the use of this product should review the latest Material Safety Data Sheet and label for product safety information, handling instructions, personal protective equipment if necessary, and any special storage condition required. The "Best use before end" date of each batch is shown on the product label. Storage beyond the date specified on the label does not necessarily mean that the product is no longer usable. In this case however, the properties required for the intended use must be checked for quality assurance reasons.

Packaging

Information on available container sizes is obtainable